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МФТИ

Методы моделирования физических процессов

Лекция №4 (МКЭ-2)

Подготовил Захаров Юрий

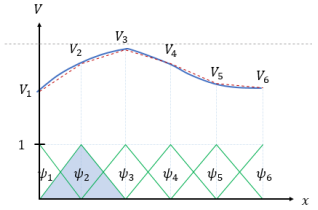
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Этапы решения задачи МКЭ

1) Сильная форма ДУ -> Слабая форма ДУ $\int_\Omega v^T A(u)\,d\Omega + \int_{\partial\Omega} \bar{v}^T B(u)\,d\Gamma = 0 \qquad \forall v,\ \bar{v}$

2) Ищем решение, как линейную комбинацию с базисными функциями



$V_N(x) = \sum_{i=1}^{2} V_i\,\psi_i(x)$

3) Подставляем базисную функцию в слабую форму $-\int_{x_1^{e1}}^{x_2^{e1}} \frac{d}{dx}\left[\sum_{i=1}^{2} V_i\,\psi_i(x)\right]\frac{dv(x)}{dx}dx = \int_{x_1^{e1}}^{x_2^{e1}}$

4) В качестве пробных функций берем базисные функции

5) Считаем матрицу жесткости одного элемента
-> решаем нелинейные уравения

$$\begin{cases} \underbrace{\left(-\int_{x_1^{e1}}^{x_2^{e1}} \frac{d\psi_1(x)}{dx}\frac{d\psi_1(x)}{dx}dx\right)}_{K_{11}} V_1^{e1} + \underbrace{\left(-\int_{x_1^{e1}}^{x_2^{e1}} \frac{d\psi_2(x)}{dx}\frac{d\psi_1(x)}{dx}dx\right)}_{K_{12}} V_2^{e1} = \underbrace{\int_{x_1^{e1}}^{x_2^{e1}} \psi_1(x) f\,dx}_{S_1} - D(x_1^{e1}) \\ \underbrace{\left(-\int_{x_1^{e1}}^{x_2^{e1}} \frac{d\psi_1(x)}{dx}\frac{d\psi_2(x)}{dx}dx\right)}_{K_{21}} V_1^{e1} + \underbrace{\left(-\int_{x_1^{e1}}^{x_2^{e1}} \frac{d\psi_2(x)}{dx}\frac{d\psi_2(x)}{dx}dx\right)}_{K_{22}} V_2^{e1} = \underbrace{\int_{x_1^{e1}}^{x_2^{e1}} \psi_2(x) f\,dx}_{S_2} + D(x_2^{e1}) \end{cases}$$

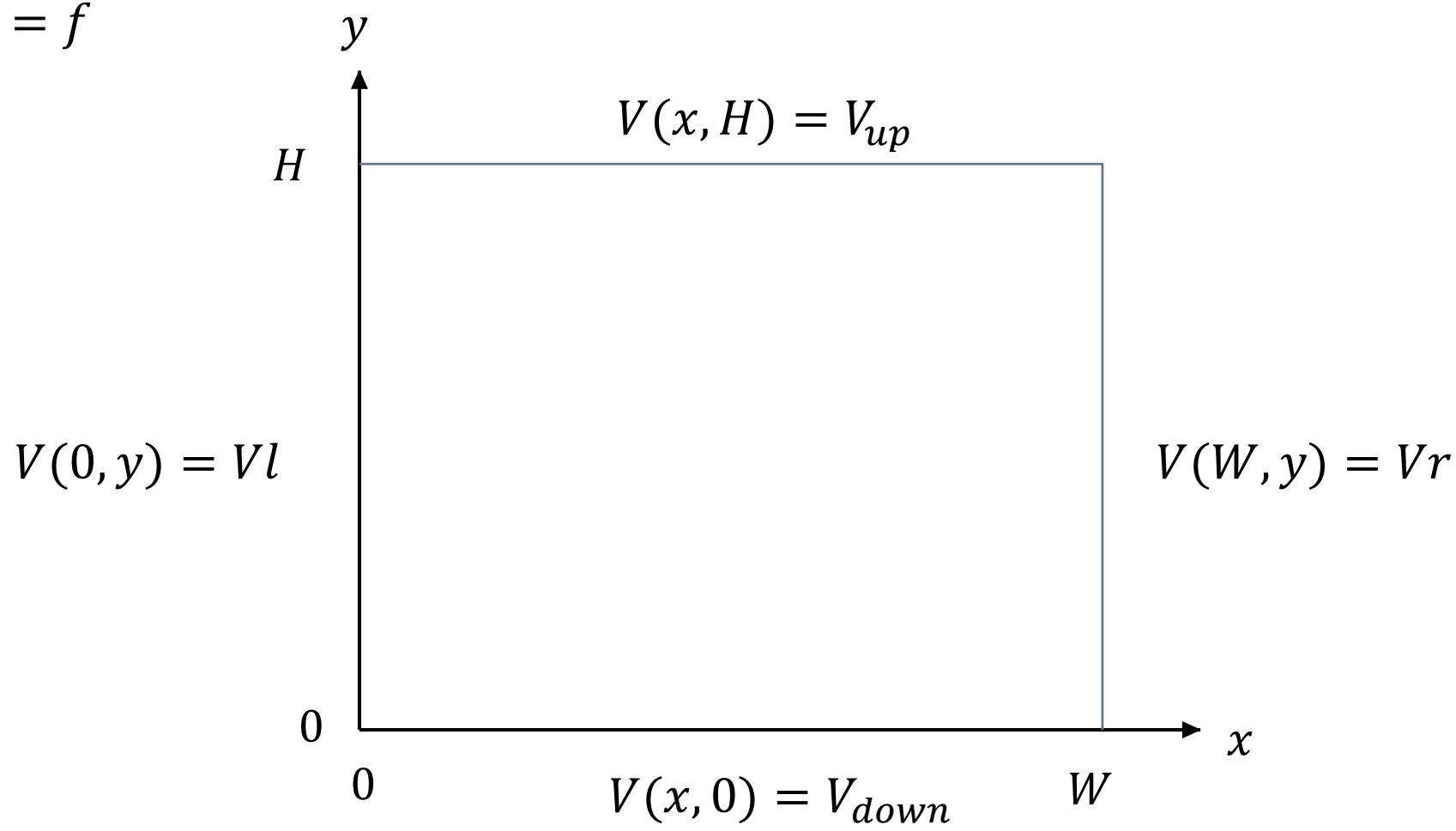
6) Собираем глобальную матрицу жесткости

$$\begin{pmatrix} 0 & -\frac{16}{3h^{e1}} & \frac{8}{3h^{e1}} & 0 & 0 & 0 & 0 \\ 0 & \frac{8}{3h^{e1}} & -\frac{7}{3h^{e1}}-\frac{7}{3h^{e2}} & \frac{8}{3h^{e2}} & -\frac{1}{3h^{e2}} & 0 & 0 \\ 0 & 0 & \frac{8}{3h^{e2}} & -\frac{16}{3h^{e2}} & \frac{8}{3h^{e2}} & 0 & 0 \\ 0 & 0 & -\frac{1}{3h^{e2}} & \frac{8}{3h^{e2}} & -\frac{7}{3h^{e2}}-\frac{7}{3h^{e3}} & \frac{8}{3h^{e3}} & 0 \\ 0 & 0 & 0 & 0 & \frac{8}{3h^{e3}} & -\frac{16}{3h^{e3}} & 0 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \\ V_6 \\ V_7 \end{pmatrix} = f \begin{pmatrix} \frac{2h^{e1}}{3} \\ \frac{h^{e1}+h^{e2}}{6} \\ \frac{2h^{e2}}{3} \\ \frac{h^{e2}+h^{e3}}{6} \\ \frac{2h^{e3}}{3} \end{pmatrix} + \begin{pmatrix} -\frac{8}{3h^{e1}}V_1 \\ \frac{1}{3h^{e1}}V_1 \\ 0 \\ \frac{1}{3h^{e3}}V_7 \\ -\frac{8}{3h^{e3}}V_7 \end{pmatrix}$$

7) Подставляем граничные условия

8) Решаем СЛАУ $\begin{bmatrix} -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} \Delta^2 f_1 - u_0 \\ \Delta^2 f_2 \\ \Delta^2 f_3 \\ \Delta^2 f_4 - u_5 \end{bmatrix}$

$$\nabla^2 V(x,y) = f$$



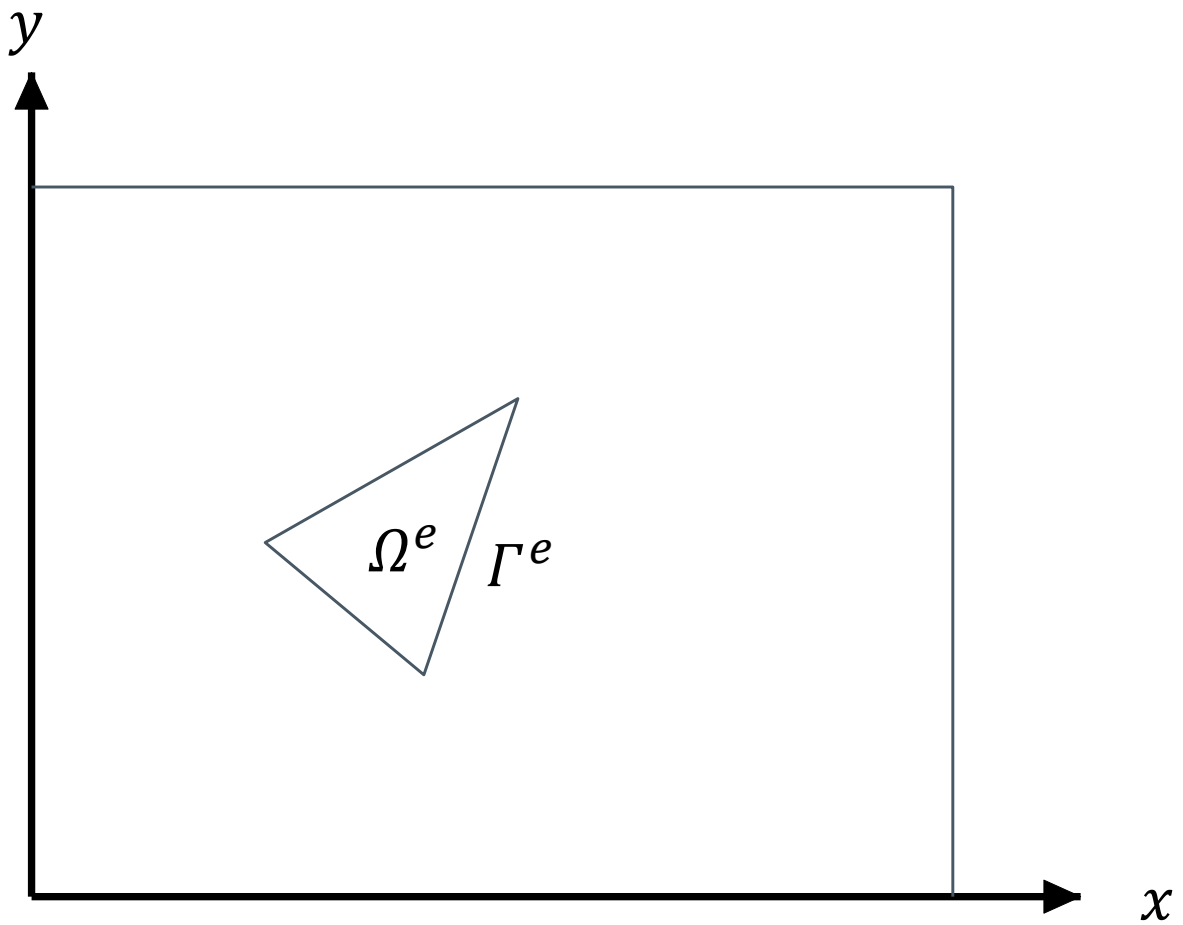
$$\nabla^2 V = f$$

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = f$$

$$\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial V}{\partial y}\right) = f$$

$$\psi\left[\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial V}{\partial y}\right)\right] = \psi f$$

$$\int_{\Omega} \psi\left[\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial V}{\partial y}\right)\right] dA = \int_{\Omega} \psi f \; dA$$



$$\int_\Omega \psi\left[\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{\partial V}{\partial y}\right)\right]dA=\int_\Omega \psi f\ \ dA$$

$$\frac{\partial}{\partial x}\left(\psi\frac{\partial V}{\partial x}\right)=\frac{\partial \psi}{\partial x}\frac{\partial V}{\partial x}+\psi\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right) \quad \Rightarrow \quad \frac{\partial}{\partial x}\left(\psi\frac{\partial V}{\partial x}\right)-\frac{\partial \psi}{\partial x}\frac{\partial V}{\partial x}=\psi\frac{\partial}{\partial x}\left(\frac{\partial V}{\partial x}\right)$$

$$\int_\Omega \left[\frac{\partial}{\partial x}\left(\psi\frac{\partial V}{\partial x}\right)-\frac{\partial \psi}{\partial x}\frac{\partial V}{\partial x}+\frac{\partial}{\partial y}\left(\psi\frac{\partial V}{\partial y}\right)-\frac{\partial \psi}{\partial y}\frac{\partial V}{\partial y}\right]dA=\int_\Omega \psi f\ \ dA$$

$$\int_\Omega \left[\frac{\partial}{\partial x}\left(\psi\frac{\partial V}{\partial x}\right)+\frac{\partial}{\partial y}\left(\psi\frac{\partial V}{\partial y}\right)\right]dA-\int_\Omega \left[\frac{\partial \psi}{\partial y}\frac{\partial V}{\partial y}+\frac{\partial \psi}{\partial x}\frac{\partial V}{\partial x}\right]dA=\int_\Omega \psi f\ \ dA$$

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$$\int_{\Omega} \left[\frac{\partial}{\partial x}\left(\psi \frac{\partial V}{\partial x}\right) + \frac{\partial}{\partial y}\left(\psi \frac{\partial V}{\partial y}\right)\right] dA - \int_{\Omega} \left[\frac{\partial \psi}{\partial x}\frac{\partial V}{\partial x} + \frac{\partial \psi}{\partial y}\frac{\partial V}{\partial y}\right] dA = \int_{\Omega} \psi f \; dA$$

$$\int_{\Omega} (\vec{\nabla} \cdot \vec{Q})\, dA = \oint_{\Gamma} \vec{Q} \cdot \hat{n} \; dl$$ Теорема грина

$$\oint_{\Gamma} \left(\left(\psi \frac{\partial V}{\partial x}\right) n_x + \left(\psi \frac{\partial V}{\partial y}\right) n_y\right) dl - \int_{\Omega} \left[\frac{\partial \psi}{\partial x}\frac{\partial V}{\partial x} + \frac{\partial \psi}{\partial y}\frac{\partial V}{\partial y}\right] dA = \int_{\Omega} \psi f \; dA$$

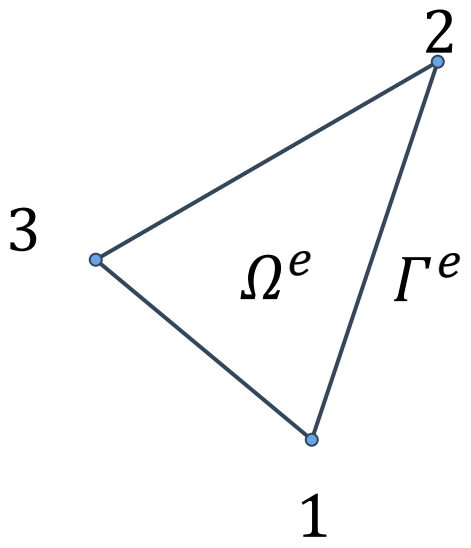
$$-\int_{\Omega} \left[\frac{\partial \psi}{\partial x}\frac{\partial V}{\partial x} + \frac{\partial \psi}{\partial y}\frac{\partial V}{\partial y}\right] dA = \int_{\Omega} \psi f \; dA - \oint_{\Gamma} \left(\left(\psi \frac{\partial V}{\partial x}\right) n_x + \left(\psi \frac{\partial V}{\partial y}\right) n_y\right) dl$$

$$-\int_{\Omega}\left[\frac{\partial\psi}{\partial x}\frac{\partial V}{\partial x}+\frac{\partial\psi}{\partial y}\frac{\partial V}{\partial y}\right]dA=\int_{\Omega}\psi f\ \ dA-\oint_{\Gamma}\left(\left(\psi\frac{\partial V}{\partial x}\right)n_x+\left(\psi\frac{\partial V}{\partial y}\right)n_y\right)dl$$

$$V=V_1^e\psi_1+V_2^e\psi_2+V_3^e\psi_3=\sum_{j=1}^{3}V_j^e\,\psi_j$$

$$-\int_{\Omega^e}\left[\frac{\partial\psi_i}{\partial x}\frac{\partial\left(\sum_{j=1}^{3}V_j^e\,\psi_j\right)}{\partial x}+\frac{\partial\psi_i}{\partial y}\frac{\partial\left(\sum_{j=1}^{3}V_j^e\,\psi_j\right)}{\partial y}\right]dA^e=$$

$$\int_{\Omega^e}\psi_i f\ \ dA^e-\oint_{\Gamma^e}\left(\left(\psi_i\frac{\partial V}{\partial x}\right)n_x+\left(\psi_i\frac{\partial V}{\partial y}\right)n_y\right)dl^e$$

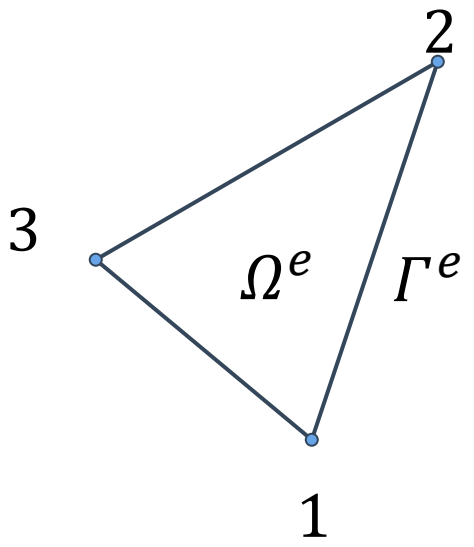


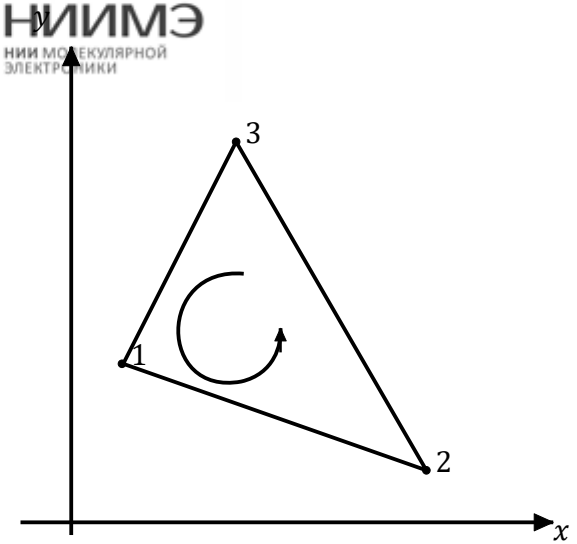
$$\begin{pmatrix} M_{11}^e & M_{12}^e & M_{13}^e \\ M_{21}^e & M_{22}^e & M_{23}^e \\ M_{31}^e & M_{32}^e & M_{33}^e \end{pmatrix} \begin{pmatrix} V_1^e \\ V_2^e \\ V_3^e \end{pmatrix} = \begin{pmatrix} S_1^e \\ S_2^e \\ S_3^e \end{pmatrix} + \begin{pmatrix} B_1^e \\ B_2^e \\ B_3^e \end{pmatrix}$$

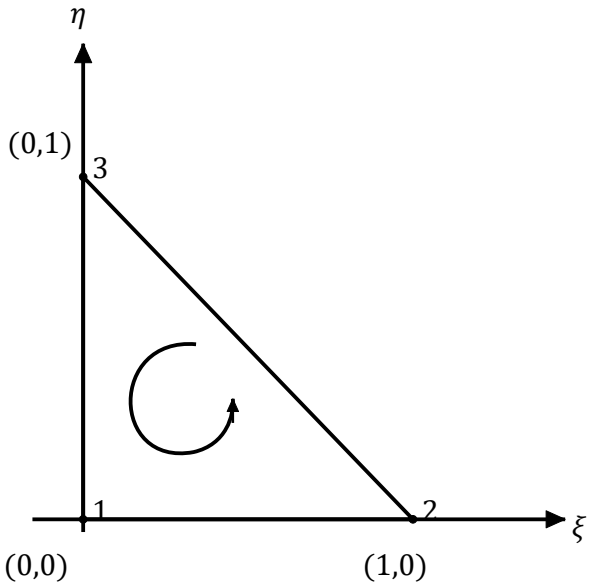
$$M_{ij}^e = -\int_{\Omega^e} \left(\frac{\partial \psi_i}{\partial x} \frac{\partial \psi_j}{\partial x} + \frac{\partial \psi_i}{\partial y} \frac{\partial \psi_j}{\partial y} \right) dA^e$$

$$S_i^e = \int_{\Omega^e} \psi_i \, f \;\; dA^e$$

$$B_1^e = -\oint_{\Gamma^e} \left(\left(\psi_i \frac{\partial V}{\partial x} \right) n_x + \left(\psi_i \frac{\partial V}{\partial y} \right) n_y \right) dl^e$$



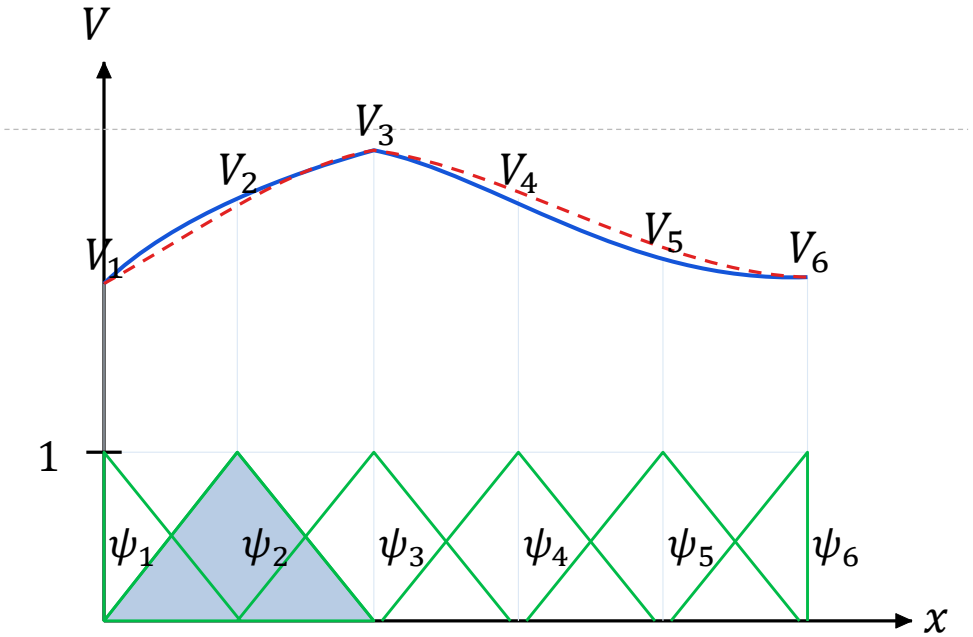


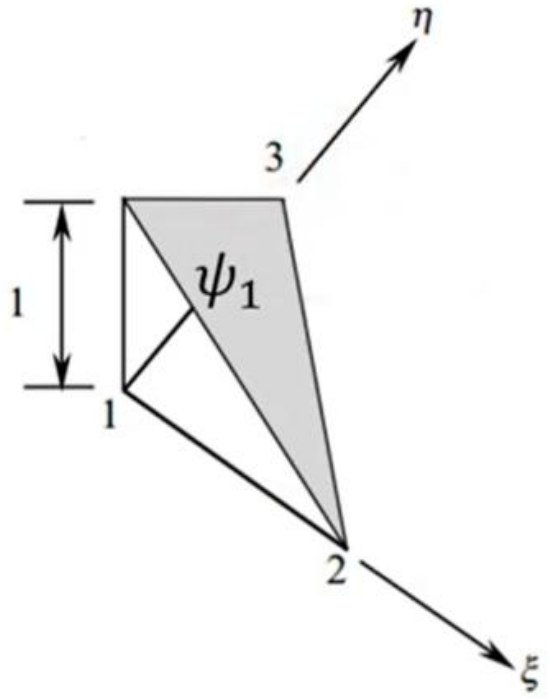


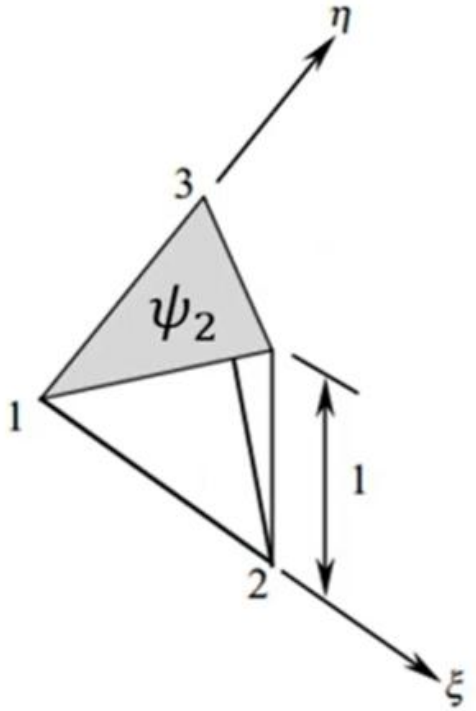
$$\psi_1(\zeta,\eta) = 1 - \zeta - \eta$$

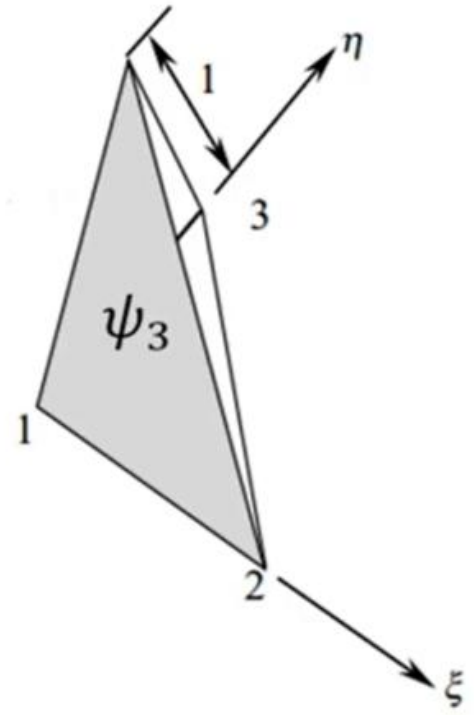
$$\psi_2(\zeta,\eta) = \zeta$$

$$\psi_3(\zeta,\eta) = \eta$$









$$\psi_1(\zeta,\eta) = 1 - \zeta - \eta$$

$$\psi_2(\zeta,\eta) = \zeta$$

$$\psi_3(\zeta,\eta) = \eta$$

$$V = V_1^e\psi_1 + V_2^e\psi_2 + V_3^e\psi_3 = \sum_{j=1}^{3} V_j^e\,\psi_j$$

$$x = x_1^e\psi_1 + x_2^e\psi_2 + x_3^e\psi_3 = \sum_{j=1}^{3} x_j^e\,\psi_j$$

$$y = y_1^e\psi_1 + y_2^e\psi_2 + y_3^e\psi_3 = \sum_{j=1}^{3} y_j^e\,\psi_j$$

$$\psi_1(\zeta,\eta) = 1 - \zeta - \eta$$

$$\psi_2(\zeta,\eta) = \zeta$$

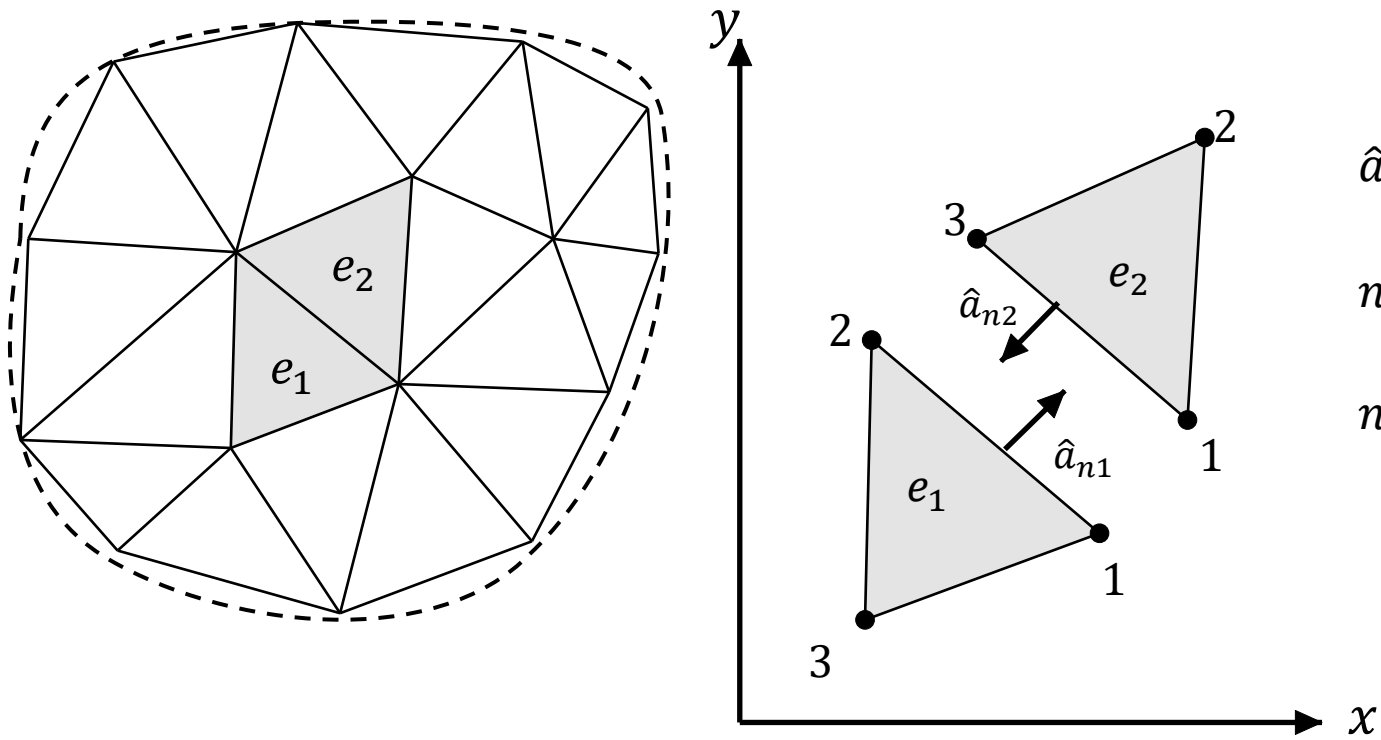
$$\psi_3(\zeta,\eta) = \eta$$

$$x = x_1^e + \overline{x}_{21}^e\zeta + \overline{x}_{31}^e\eta$$

$$y = y_1^e + \overline{y}_{21}^e\zeta + \overline{y}_{31}^e\eta$$

$$\begin{bmatrix} M_{11}^e = -\left[\frac{\overline{y}_{23}^2}{4A^e} + \frac{\overline{x}_{32}^2}{4A^e}\right] & M_{12}^e = -\left[\frac{\overline{y}_{23}\overline{y}_{31}}{4A^e} + \frac{\overline{x}_{32}\overline{x}_{13}}{4A^e}\right] & M_{13}^e = -\left[\frac{\overline{y}_{23}\overline{y}_{12}}{4A^e} + \frac{\overline{x}_{32}\overline{x}_{21}}{4A^e}\right] \\ M_{21}^e = -\left[\frac{\overline{y}_{23}\overline{y}_{31}}{4A^e} + \frac{\overline{x}_{32}\overline{x}_{13}}{4A^e}\right] & M_{22}^e = -\left[\frac{\overline{y}_{31}^2}{4A^e} + \frac{\overline{x}_{13}^2}{4A^e}\right] & M_{23}^e = -\left[\frac{\overline{y}_{31}\overline{y}_{12}}{4A^e} + \frac{\overline{x}_{13}\overline{x}_{21}}{4A^e}\right] \\ M_{31}^e = -\left[\frac{\overline{y}_{23}\overline{y}_{12}}{4A^e} + \frac{\overline{x}_{32}\overline{x}_{21}}{4A^e}\right] & M_{32}^e = -\left[\frac{\overline{y}_{31}\overline{y}_{12}}{4A^e} + \frac{\overline{x}_{13}\overline{x}_{21}}{4A^e}\right] & M_{33}^e = -\left[\frac{\overline{y}_{12}^2}{4A^e} + \frac{\overline{x}_{21}^2}{4A^e}\right] \end{bmatrix}$$

$$S = \frac{f A^e}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$



$$\hat{a}_{n1} = -\hat{a}_{n2}$$

$$n_x^{e1} = -n_x^{e2}$$

$$n_y^{e1} = -n_y^{e2}$$

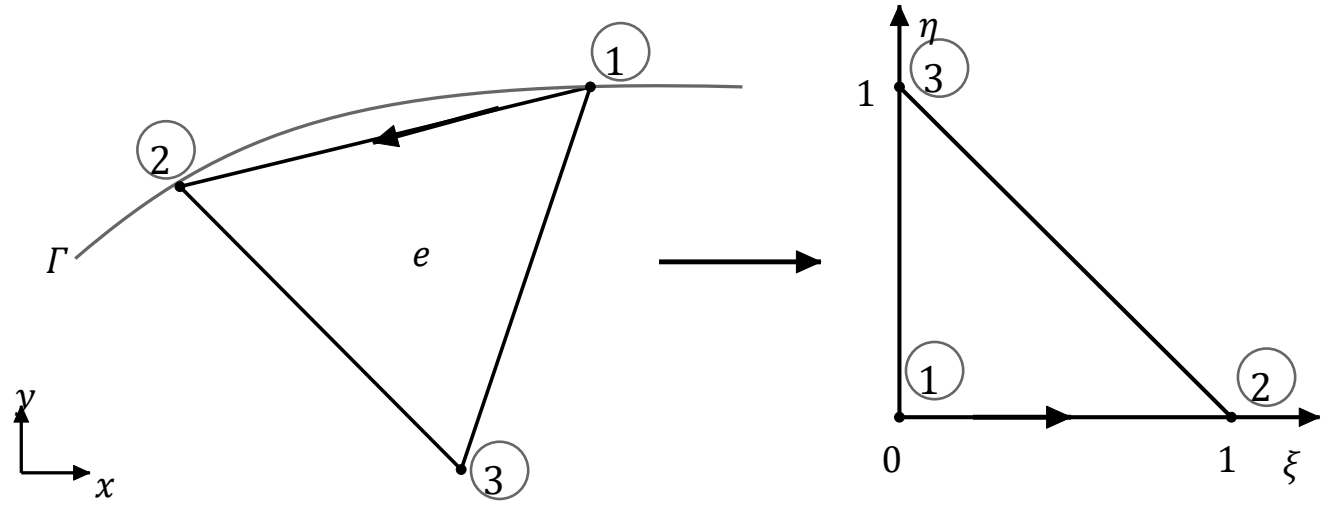
$$B_1^{e1} = -\int_{1\to2} \left(\left(\psi_1^{e1} \frac{\partial V}{\partial x} \right) n_x^{e1} + \left(\psi_1^{e1} \frac{\partial V}{\partial y} \right) n_y^{e1} \right) dl^{e1}$$

$$B_1^{e2} = -\int_{2\to1} \left(\left(\psi_1^{e2} \frac{\partial V}{\partial x} \right) n_x^{e2} + \left(\psi_1^{e2} \frac{\partial V}{\partial y} \right) n_y^{e2} \right) dl^{e2}$$

$$B_1^{e1} + B_1^{e2} = 0$$

$$B_i = -\int_{1\to2} q\,\psi_i\ dl + \int_{1\to2} \gamma\,\psi_i(V_1^e\psi_1 + V_2^e\psi_2 + V_3^e\psi_3)dl$$

$$B = -\frac{q l_{12}}{2}\begin{pmatrix}1\\1\\0\end{pmatrix}$$



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$$S = \frac{fA^e}{3}\begin{pmatrix}1\\1\\1\end{pmatrix} \qquad B = -\frac{ql_{12}}{2}\begin{pmatrix}1\\1\\0\end{pmatrix}$$

$$M_{11}^e = M_{11}^e - \frac{\gamma l_{12}}{3} \qquad M_{21}^e = M_{21}^e - \frac{\gamma l_{12}}{6} \qquad M_{21}^e = M_{21}^e - \frac{\gamma l_{12}}{6}$$

$$M_{12}^e = M_{12}^e - \frac{\gamma l_{12}}{6} \qquad M_{22}^e = M_{22}^e - \frac{\gamma l_{12}}{3} \qquad M_{22}^e = M_{22}^e - \frac{\gamma l_{12}}{3}$$

$$M_{13}^e = M_{13}^e - 0 \qquad M_{23}^e = M_{23}^e - 0 \qquad M_{23}^e = M_{23}^e - 0$$

Виды граничных условий

$V = Fixed\ value$

$\nabla . \boldsymbol{V} = Fixed\ value$

$\nabla . \boldsymbol{V} + \gamma V = Fixed\ value$

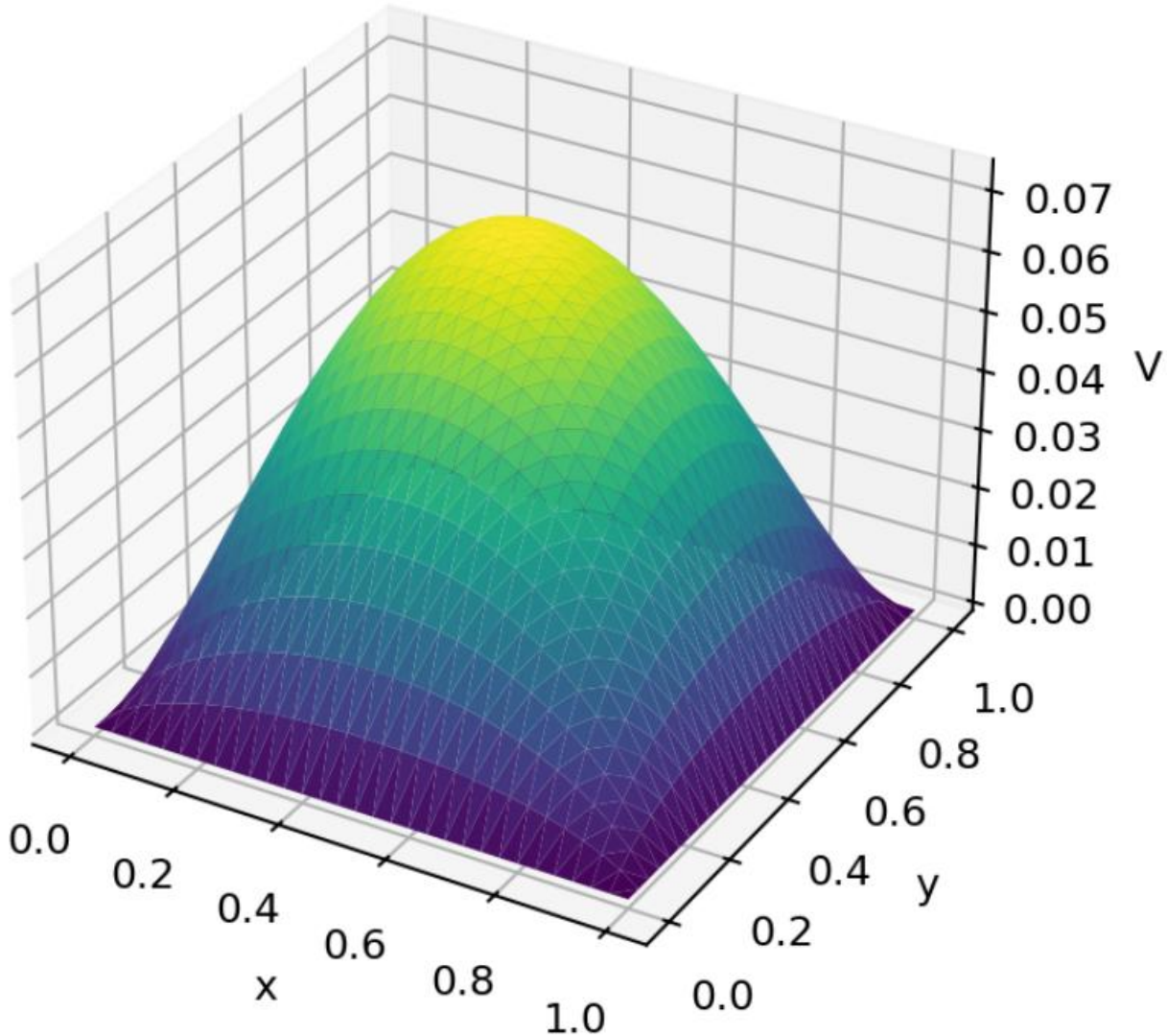
$$\begin{pmatrix} M_{11}^e & M_{12}^e & M_{13}^e \\ M_{21}^e & M_{22}^e & M_{23}^e \\ M_{31}^e & M_{32}^e & M_{33}^e \end{pmatrix} \begin{pmatrix} V_1^e \\ V_2^e \\ V_3^e \end{pmatrix} = \begin{pmatrix} S_1^e \\ S_2^e \\ S_3^e \end{pmatrix} + \begin{pmatrix} B_1^e \\ B_2^e \\ B_3^e \end{pmatrix}$$

$$\mathbf{M}_e\mathbf{V} = \mathbf{S} + \mathbf{B}$$

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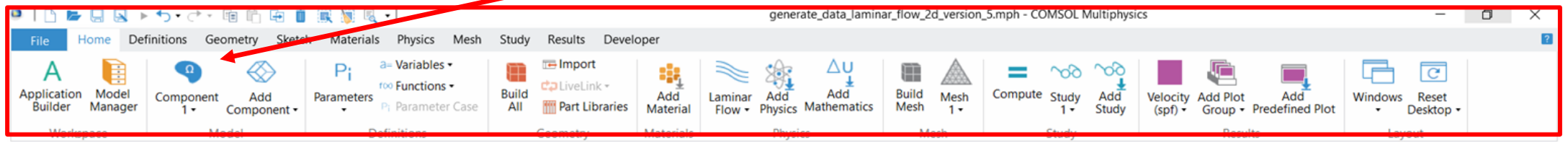
$\Delta V = -1$, $V = 0$ на границе



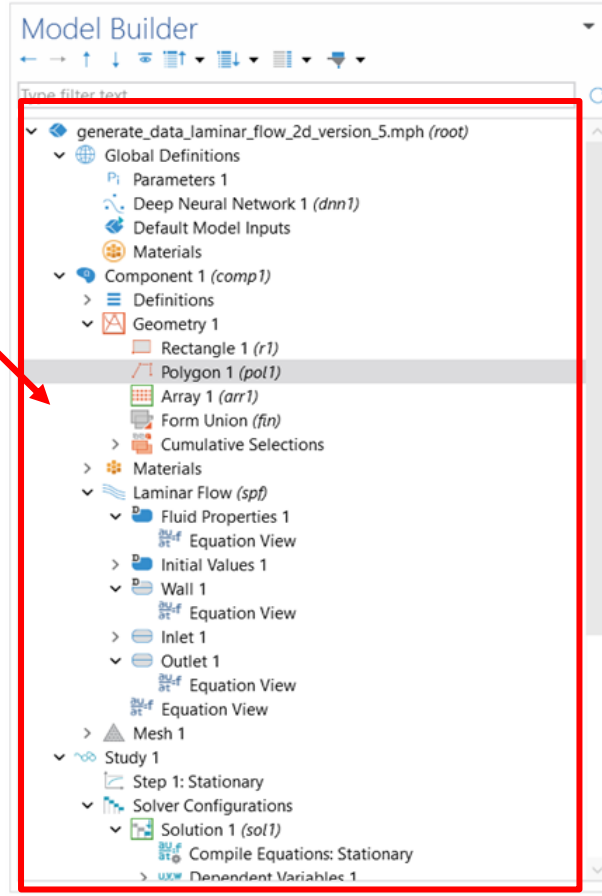


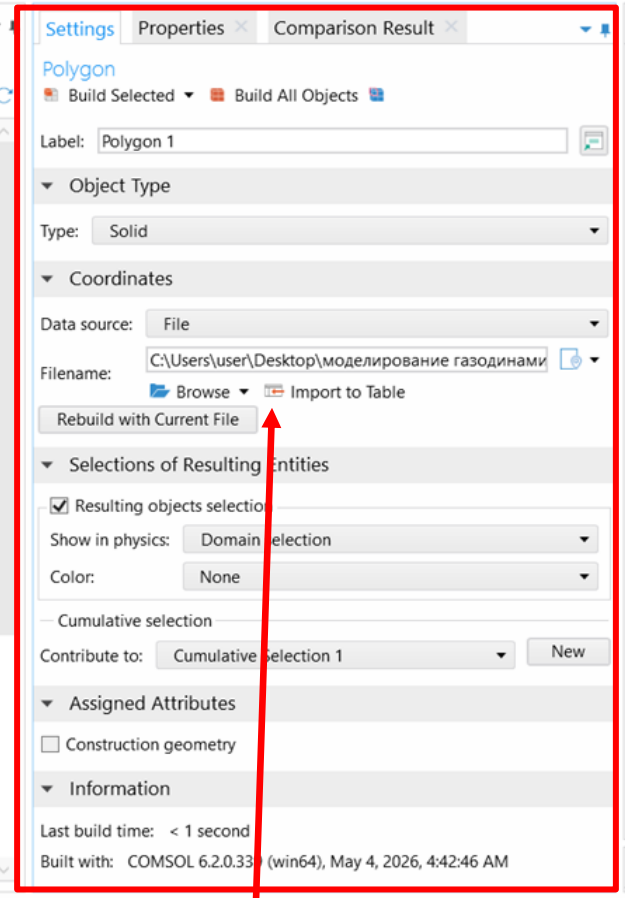
- Реализован набор встроенных математических моделей физических процессов
- Редактор геометрических моделей
- Сеточный генератор
- Набор решателей ДУ
- Встроенные элементы визуализации
- Встроенная структура нейронных сетей для создания суррогатных моделей
- Среда разработки приложений

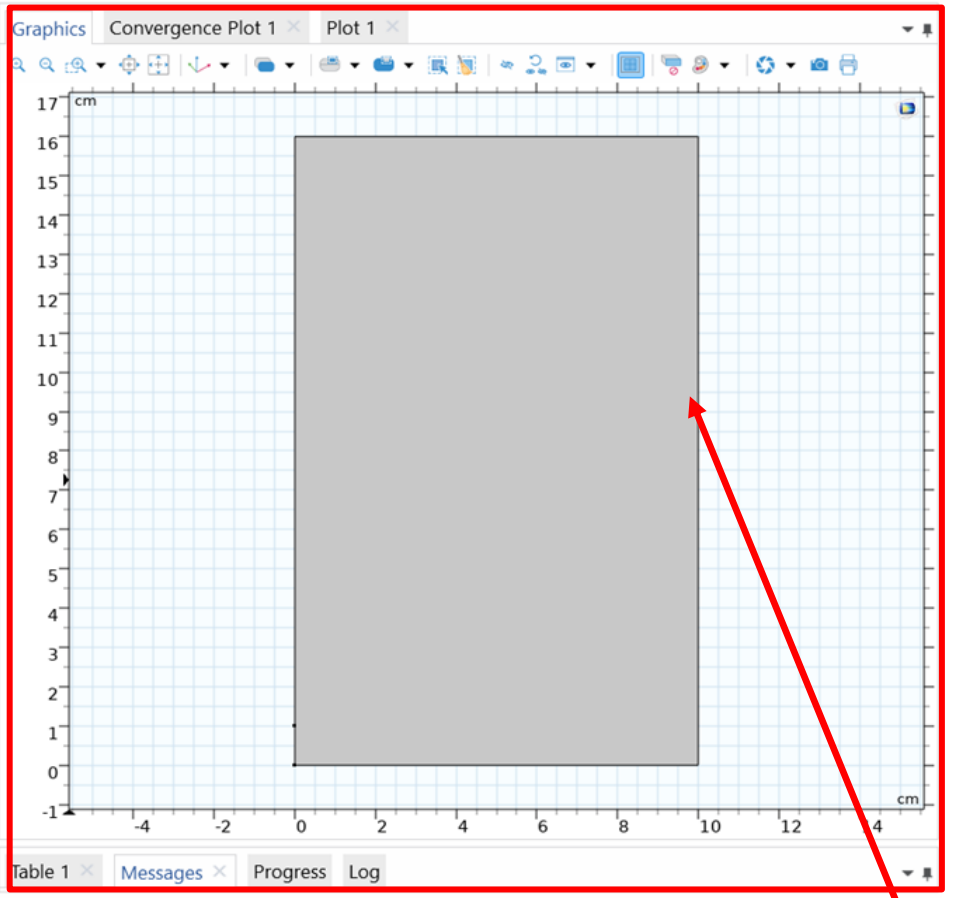
панель инструментов



Дерево модели







1.85 GB | 2 GB

Окно настройки

отображение объекта